



Slowave Core Algorithm — Overview

Architecture

Slowave is a **dual-layer memory system** for AI agents. It combines:

1. **Latent Layer** (SlowWave substrate): Continuous vector embeddings processed via FAISS, prototype clustering, graph propagation, and temporal encoding. Brain-inspired: hippocampus analogue.
2. **Symbolic Layer** (Slowave additions): Discrete text schemas, FTS indexes, raw event logs, sessions. Brain-inspired: cortical consolidation analogue.

remember(content)

← Agent writes

INGESTION

text → encode() → episodic memory (FAISS+SQLite)
salience = compute_novelty(nn_similarity)

REPLAY (periodic / session-end)

sample episodes → assign to prototypes
update graph edges (similarity + transition)
self-supervised rehearsal

CONSOLIDATION

prototypes → latent schemas → SQL schemas
supersession detection / contradiction

RETRIEVAL

query → FAISS cosine + spreading activation
→ temporal boost → salience re-rank → top-k

CONTEXT GATING

schemas → keyword scoring → activation ranking
→ MMR dedup → budget trim → rendered text

FEEDBACK LOOP

user feedback → FeedbackSignal → schema updates
salience_delta, confidence_delta, review flags

Data Model

Core Types

Type	Description	Key Fields
Event	Raw input envelope	event_id, timestamp, type, embedding
EpisodicMemory	Stored vector trace	id, embedding, salience, recalled_count
SemanticPrototype	Cluster centroid	id, centroid, support_count, variance, scale
Schema	Symbolic memory record	id, content_text, embedding, facets, status
RetrievedMemorySet	Recall result	query_embedding, episodic, prototypes, neighbors

Scales (Stage 9)

Two hippocampal subfield analogues:

- fine** (CA3): High assignment threshold — pattern completion. Distinct but related facts stay separate.
- coarse** (CA1): Low assignment threshold — pattern generalization. Similar episodes merge into broader concepts.

Global Constants (Dimension)

Constant	Value	Description
Default dim	384	Embedding dimension

Constant	Value	Description
		(paraphrase-multilingual-MiniLM-L12-v2)
Default model	<code>sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2</code>	ONNX Runtime backend

Files Structure

File	Module	Purpose
<code>slowave/core/engine.py</code>	<code>SlowaveEngine</code>	Top-level facade: remember, recall, consolidate
<code>slowave/core/config.py</code>	<code>SlowaveConfig</code>	Root configuration dataclass
<code>slowave/latent/episodic_store.py</code>	<code>EpisodicStore</code>	FAISS-backed episodic memory
<code>slowave/latent/semantic_store.py</code>	<code>SemanticStore</code>	FAISS-backed prototype store
<code>slowave/latent/salience.py</code>	<code>SalienceEngine</code>	Novelty, decay,

File	Module	Purpose
		reinforcement
<code>slowave/latent/replay_engine.py</code>	<code>ReplayEngine</code>	Prototype assignment + graph building
<code>slowave/latent/graph_manager.py</code>	<code>GraphManager</code>	Weighted directed prototype edges
<code>slowave/latent/retrieval.py</code>	<code>RetrievalPipeline</code>	Cosine + spreading activation
<code>slowave/latent/temporal.py</code>	<code>TemporalProbe</code>	Multi-scale sinusoidal temporal encoding
<code>slowave/latent/transition_model.py</code>	<code>TransitionModel</code>	Graph-based successor prediction
<code>slowave/latent/vsa.py</code>	VSA bind/unbind	HRR-based role binding (experimental)
<code>slowave/core/consolidation.py</code>	<code>Consolidator</code>	Prototype → schema conversion
<code>slowave/core/context.py</code>	<code>WorkingMemoryGate</code>	Keyword-based activation gating

File	Module	Purpose
<code>slowave/core/feedback.py</code>	Feedback system	Learning signal computation
<code>slowave/core/supersession_manifold.py</code>	<code>SupersessionManifold</code>	SVD1 direction for update detection